

Faculty of Computing



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SAP 64281

[Computer Communications & Network]

Lab No 5 Tasks

Task 1: Use the IP address chart and your knowledge of IP address classes to answer the following questions:

1. **What is the decimal and binary range of the first octet of all possible Class B IP addresses?**

☐ Decimal: From: 128_____ To: 191

☐ Binary: From: 10000000_____ To: _____ 10111111

2. **Which octet(s) represent the network portion of a Class C IP address?**

1ST, 2ND, 3RD

3. **Which octet(s) represent the host portion of a Class A IP address?**

2ND, 3RD, 4TH

4. **What is the maximum number of useable hosts with a Class C network address?**

254

$2^8 - 2 = 254$ usable host

5. **How many Class B networks are there?**

16,382

$2^{14} - 2 = 16,384 - 2 = 16,382$

6. **How many hosts can each Class B network have?**

65,534

$2^{16} - 2 = 65,536 - 2 = 65,534$

7. **How many octets are there in an IP address? 4 How many bits per octet? 8**

Task 2: Determine the host and network portions of the IP address

With the following IP host addresses, indicate the following:

- ☐ Class of each address
- ☐ Network address or ID
- ☐ Host portion
- ☐ Default subnet mask

The host portion will be all zeros for the network ID. Enter just the octets that make up the host. The host portion will be all ones for a broadcast. The network portion of the address will be all ones for the subnet mask. Fill in the following table:

Host IP Address	Address Class	Network Address	Host Address	Network Broadcast Address	Default Subnet Mask
216.14.55.137	Class C	216.14.55.0	0.0.0.137	216.14.55.255	255.255.255.0
123.1.1.15	Class A	123.0.0.0	0.1.1.15	123.255.255.255	255.0.0.0
150.127.221.244	Class B	150.127.0.0	0.0.221.244	150.127.255.255	255.255.0.0
194.125.35.199	Class C	194.125.35.0	0.0.0.199	194.125.35.255	255.255.255.0
175.12.239.244	Class B	175.12.0.0	0.0.239.244	175.12.255.255	255.255.0.0

Task 3: Given an IP address of 142.226.0.15, answer the following questions:

What is the binary equivalent of the second octet? 11100010

What is the class of the address?

Class B

First octet is 142, which falls in 128–191 (Class B)

What is the network address of this IP address? 142.226.0.0 , Default Class B subnet mask: 255.255.0.0

Is this a valid IP host address (Y/N)? Why or why not?

Yes: because It's **not** the network address (not ending in .0.0), **not** the broadcast address (.255.255), and so it is belongs to valid range.

